COPYRIGHT RESERVED BCA (Sem-IV) – Database Management System (403) Core-X

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

   (a) What is a database?

(i) Organized collection of information that cannot be accessed, updated, and managed.

(ii) Collection of data or information without organizing.

(iii) Organized collection of data or information that can be accessed, updated, and managed.

(iv) Organized collection of data that cannot be updated.

(b) The ability to query data, as well as insert, delete, and alter tuples, is offered by ____

(i) TCL (Transaction Control Language)

(ii) DCL (Data Control Language)

(iii) DDL (Data Definition Language)

(iv) DML (Data Manipulation Language)

(c) Why the following statement is erroneous?

```
SELECT dept_name, ID, avg (salary)
FROM instructor
GROUP BY dept_name;
```

(i) Dept_id should not be used in group by clause

(ii) Group by clause is not valid in this query

(iii) Avg (salary) should not be selected

(iv) None

(d) After groups have been established, SQL applies predicates in the _____ clause, allowing aggregate functions to be used.

i) Where

ii) Having

iii) Group by

iv) With

(e) For designing a normal RDBMS which of the following normal form is considered adequate?

i) 4NF

ii) 3NF

iii) 2NF

iv) 5NF

(f) What is a primary key used for?

i) Uniquely identify a record in a table

ii) Enforce relationships between tables

iii) Prevent data duplication

v) All of the above

(g) Which of the following is NOT a type of relationship in a relational database?

i) One-to-One

ii) One-to-Many

iii) Many-to-M

iv) Many-to-One

(h) Which command is used to remove a table from a database in SQL?

i) DELETE

ii) DROP

iii) REMOVE

iv) TRUNCATE

(i) What is a foreign key in a relational database?

i) A key used to identify a column uniquely in a table

ii) A key used to connect two tables

iii) A key that does not allow NULL values

iv) A key used to store passwords

(j) Which of the following establishes a top-to-bottom relationship among the items?

i) Relational schema

ii) Network schema

iii) Hierarchical schema

iv) All of the mentioned

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Explain DDL and DML commands with Example.
3. Draw and Explain E-R model Symbol.
4. Give the Difference between primary key and unique key.
5. Differentiate between DBMS and File management system.
6. Define E-R model with examples.
7. Explain Generalization and Specialization with example.

8. What do you understand by Data Constraints in SQL? Explain various types with the help of examples.

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

9. Write short notes on:
   (i) CHECK constraint.
   (ii) 3rd NF and BCNF
   (iii) Inner Join
10. What do you mean by operator? What are the different types of operators in SQL?
11. Explain DML Statements with syntax and example and also define group function with syntax :

    AVG(), SUM(), MIN(), MAX() and COUNT().

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